DTU PanRes: A structure-guided resource bridging acquired and latent antimicrobial resistance

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1. INTRODUCTION

Antimicrobial resistance (AMR) genes are spread across resources with **different scopes and annotations** that are hard to combine.

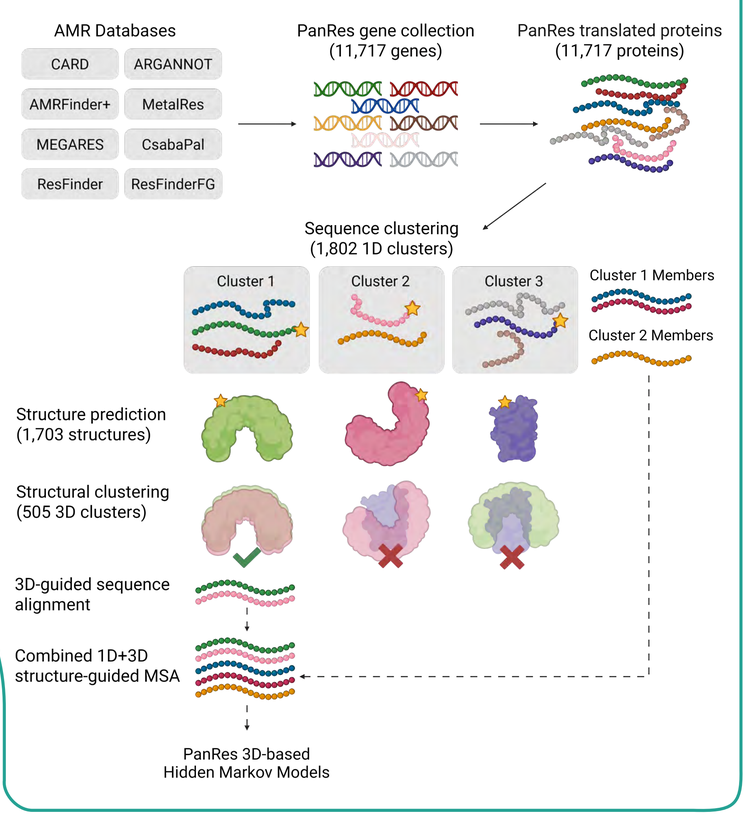
Latent resistance determinants discovered by functional metagenomics remain **poorly connected** to clinically recognised genes.

PanRes links genes, proteins, structures, ontology and 3D-HMMs to enable sensitive **detection beyond close sequence matches.**

2. BUILDING THE DATABASE

Resistance genes were collected across databases, translated, modelled as protein structures, and grouped into structural clusters. Structure-guided alignments were then used to build **3D-HMMs.**

PanRes Ontology connects genes, proteins, structures, clusters, and metadata in a unified framework, accessible through a **web interface.**

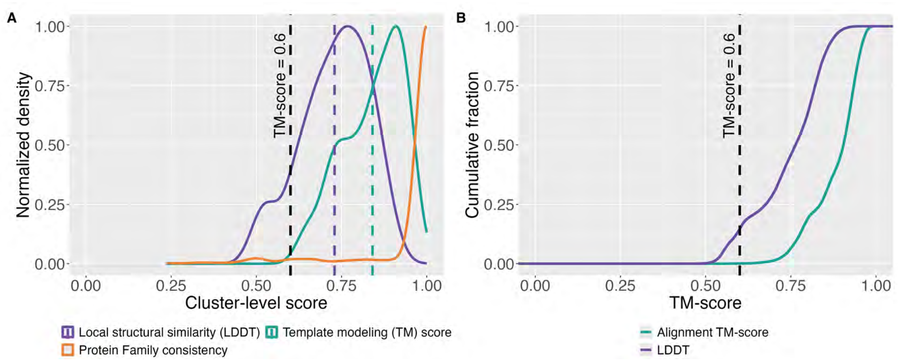


SEQUENCE DIVERSITY

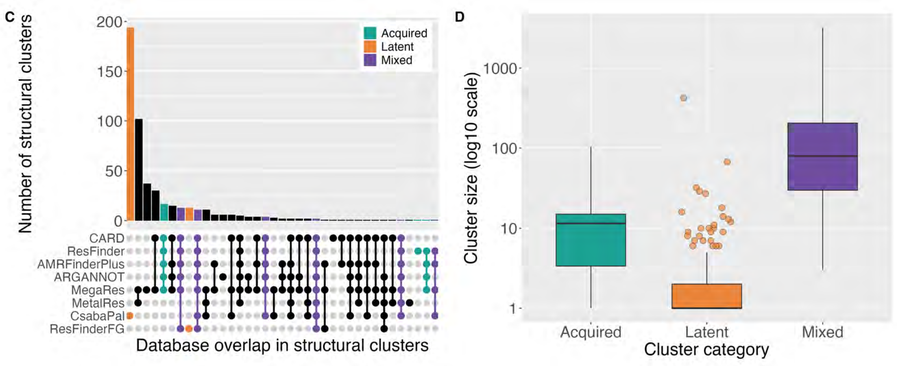
Genes - Proteins - Annotations



3. PANRES STRUCTURAL RESISTOME



Metrics describing structural agreement were used to assess cluster quality. Clusters were structurally coherent as a whole (**A**), while pairwise structure alignments confirmed strong agreement between cluster members, with **95%** of TM-scores exceeding 0.6, consistent with a shared fold (**B**).

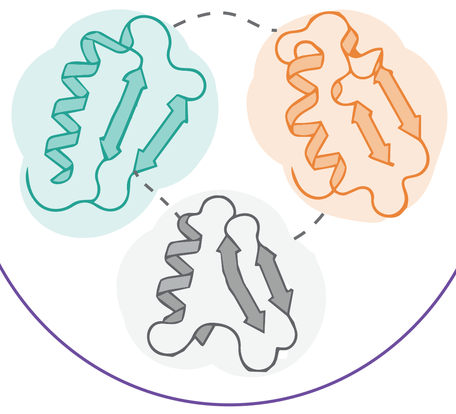


Structural clustering showed limited overlap between acquired and latent resistance (**C**):

- **194** clusters contained **only latent** proteins
- **45** clusters were **mixed**

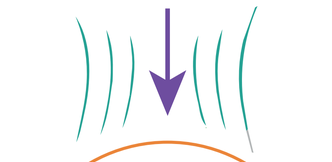
Latent-only clusters mostly remained singletons (**D**), showing how undersampled the latent resistome is.

PanRes

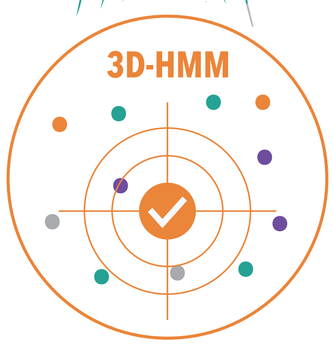


STRUCTURAL INSIGHT

Prediction - Clustering - Alignment



3D-HMM

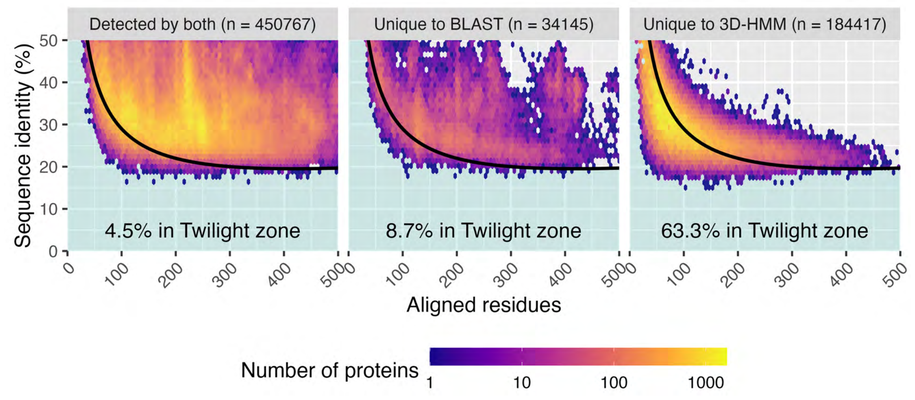


AMR DISCOVERY

Detection of distant resistance

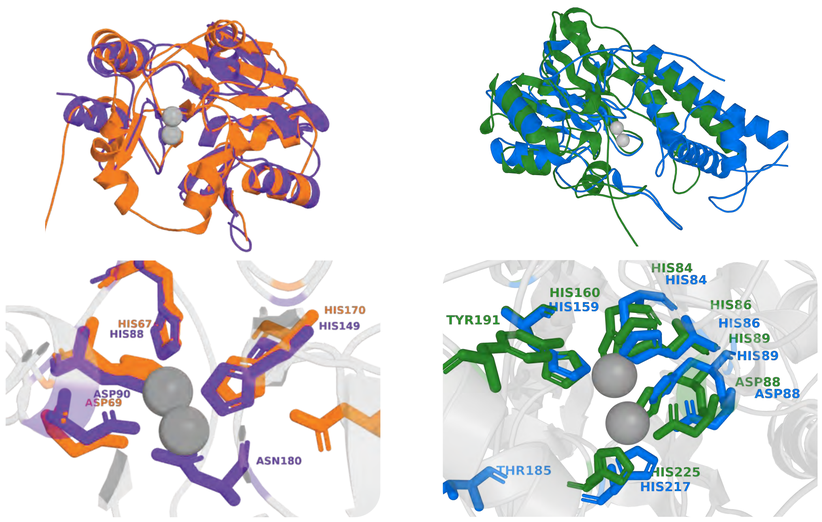
4. NOVEL β-LACTAMASE CANDIDATES

Structure-guided HMMs extend detection beyond the conventional sequence "twilight zone" and identify more distant homologs compared to BLASTp.



3D-HMMs detected distant β-lactamase candidates. Catalytic residues were used to prioritise the hits.

Despite low sequence similarity, distant candidates retained β-lactamase-like folds and catalytic sites.



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